

Sam Njoroge

Software Engineer | Platform Engineering | Hardware-Software Integration

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SUMMARY

Software engineer with 10+ years optimizing integrated platforms across embedded devices, APIs, and cloud infrastructure. Expertise in end-to-end system debugging, performance optimization at scale, and architecting resilient distributed systems. Ability to bridge technical domains, from firmware and hardware integration to micro services and cloud-native architectures, delivering solutions that serve millions of users with high reliability..

I hold a bachelor's degree in information technology from the University of Massachusetts, where I acquired strong knowledge and skills in data structures, algorithms, databases, and software engineering. I am passionate about learning new technologies and solving complex problems. I am always eager to take on new challenges and contribute to the success of the organization.

CORE COMPETENCIES

Software Engineer: End-to-end platform design, distributed systems, micro services, embedded Linux systems

Integration Expertise: Hardware-software integration, chipset collaboration, firmware deployment, API design

Performance & Reliability: System optimization, debugging complex failures, observability, 99.99%+ uptime

Technical Domains: Embedded platforms, streaming infrastructure, authentication/authorization, IoT devices

Infrastructure: Linux/Unix systems, containerization, cloud platforms (AWS/GCP), CI/CD pipelines

PROFESSIONAL EXPERIENCE

Technical Lead – Fios Streaming Platform Architecture

HCL America | January 2022 – Present | Irving, TX

Architected system improvements for Verizon's multi-platform streaming ecosystem spanning mobile apps, set-top boxes, and backend services serving 5M+ monthly active users.

Platform Architecture:

- Designed self-healing mechanisms and diagnostic frameworks that reduced production incidents by 97% (6,000 to 200 weekly)
- Architected cross-platform feature parity between mobile applications and embedded set-top box devices
- Established observability architecture for real-time monitoring across mobile, embedded, and cloud layers

System Integration:

- Bridged technical requirements across iOS/Android mobile apps, embedded Linux set-top boxes, and cloud API infrastructure
- Designed data flow architectures ensuring consistent user experience across heterogeneous device ecosystem
- Collaborated with hardware vendors, mobile teams, and backend services to architect integrated solutions

Technical Impact:

- Transformed reactive incident response into proactive system monitoring with predictive failure detection
- Architected metric collection framework providing actionable insights across entire platform stack
- Designed backward-compatible API changes supporting seamless migration to next-generation platform

Senior Software Engineer - Authentication & Authorization Architecture

HCL America | August 2019 – January 2022 | Irving, TX

Implemented high-performance authentication infrastructure serving Verizon's streaming services.

System Design:

- Implemented secure API gateway handling millions of authorizations per minute with 99.99% uptime
- Designed OAuth 2.0, JWT, and SSO integration patterns for secure multi-platform authentication
- Implemented distributed caching architecture reducing authorization latency by 60%
- Designed database optimization strategies for high-throughput entitlement queries

Performance Optimization:

- Implemented asynchronous processing patterns improving API response times under peak load
- Designed auto-scaling infrastructure responding to dynamic traffic patterns
- Implemented comprehensive observability stack (Splunk, Prometheus, Grafana) for performance analysis

Reliability Engineering:

- Designed fault-tolerant systems with graceful degradation under partial outages
- Architected circuit breaker patterns and retry logic for external service dependencies
- Implemented comprehensive logging and tracing for distributed request debugging

Software Engineer - Embedded Platform Architecture

Tekmark & HCL America (Verizon) | October 2015 – August 2019 | Irving, TX

Developed embedded Linux platforms for Wi-Fi routers and TV streaming set-top boxes.

Embedded Systems Architecture:

- Developed zero-touch provisioning system automating device initialization and configuration
- Designed firmware deployment pipeline supporting multiple hardware platforms with version compatibility
- Implemented boot sequence optimization and failure recovery mechanisms for embedded devices

Hardware-Software Integration:

- Collaborated with chipset vendors (Broadcom, Qualcomm) to architect MAC layer implementations for Wi-Fi standards
- Designed hardware abstraction layers enabling firmware portability across device generations
- Implemented logging infrastructure for debugging distributed embedded device deployments

Platform Debugging:

- Debugged complex boot failures, kernel panics, and hardware-software interface issues
- Designed diagnostic tools for remote troubleshooting of deployed customer devices
- Implemented comprehensive logging framework for production issue investigation

System Integration:

- Integrated CESDK for automated device discovery and provisioning workflows
- Designed component integration architecture for zero-touch deployment reducing manual intervention by 80%
- Implemented environmental monitoring and failure detection systems

Unix Systems Engineer - Infrastructure Architecture

TSP (Texas Instruments) | March 2014 – August 2015 | Dallas, TX

Managed Unix/Linux infrastructure for Fortune 500 semiconductor manufacturing operations.

Infrastructure Design:

- Implemented monitoring infrastructure across thousands of Unix/Linux servers ensuring 24/7 production uptime
- Designed automated backup and recovery architecture using VERITAS NetBackup
- Implemented SAN and NAS storage architectures optimizing data access patterns

System Automation:

- Developed Bash and Python automation frameworks reducing manual operational overhead
- Designed patrol monitoring agent deployment for enterprise-scale server infrastructure

OPEN SOURCE CONTRIBUTIONS

Creator & Lead Developer - Multi-Agent AI Communication Platform

GitHub: <https://github.com/A2AGateway> | **2025 – Present**

Designed and developed an open source communication protocol and gateway platform enabling autonomous AI agents to discover, coordinate, and collaborate with each other in distributed multi-agent systems.

- Architected Agent-to-Agent (A2A) protocol implementation in Go, establishing a standardized framework for AI agents to communicate, delegate tasks, and share knowledge across platforms and organizations.
- Built core agentic infrastructure including dynamic agent discovery and registration, capability-based routing, structured task delegation between agents, and real-time event notifications for agent coordination.
- Implemented robust message handling and error recovery for reliable inter-agent communication, ensuring agents can autonomously retry, fallback, and recover from failures in distributed environments.

TECHNICAL SKILLS

Languages: C/C++, Go, Python, Java, Bash

Operating Systems: Linux (RHEL, CentOS, Ubuntu), Unix (Solaris), Embedded Linux

Architecture Patterns: Microservices, event-driven, distributed systems, embedded platforms

Infrastructure: Docker, Kubernetes, Jenkins, Ansible, Terraform, AWS, GCP **Databases:** MySQL, PostgreSQL, MongoDB, Cassandra

Monitoring & Debugging: Splunk, Prometheus, Grafana, ELK Stack, system-level debugging

Security: OAuth 2.0, JWT, SSO, API security, firmware security

Hardware Integration: Chipset collaboration, firmware deployment, MAC layer implementation

EDUCATION

Bachelor of Science in Information Technology University of Massachusetts | 2014 – 2019

Self-Driving Car Nanodegree Udacity | 2017 – 2018

KEY PROJECTS & ACHIEVEMENTS

High-Performance API Gateway: Architected secure gateway handling 1M+ requests/minute with 99.99% uptime

Zero-Touch Provisioning Platform: Designed embedded system reducing deployment time by 80%

Self-Healing Architecture: Designed diagnostic framework reducing production incidents by 97%

A2A Protocol Gateway: Open-source distributed systems protocol for autonomous agent communication

Machine Learning Integration: Implemented TensorFlow-based traffic sign classification (98% accuracy) and lane detection for autonomous vehicle systems